

Level and Shape: Learning Adjacent-Lanthanide Selectivity from the Solvent-Extraction Record

Bogdan Mironov

Department of Chemistry, Berea College
Berea, KY, USA

mironovb@berea.edu Dias Daulet
NIS Nauryzbay

Almaty, Kazakhstan

daulet_d0806@hbalm.nis.edu.kz Konstantinos D. Vogiatzis

Department of Chemistry, University of Tennessee
Knoxville, TN, USA

kvogiatz@utk.edu

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Abstract

Adjacent lanthanide ions differ in radius by about 0.013 Å, and separating them is the most expensive step in rare-earth processing. In this work, we show that the published solvent-extraction record contains sufficient information to learn which ligands perform this separation effectively. On 4,746 measurements covering 162 extractants, a gradient-boosting model evaluated with whole extractants held out predicts log separation factors that correlate with experiment at $r = 0.59$ across 905 adjacent pairs. The same data give $R^2 = 0.85$ for the distribution ratio under the random row split of previous work and 0.49 with extractants held out; adjacent selectivity is the 12 % of the variance that remains once the level of each series is removed, and it is scored on that part alone. The prediction consists of two parts: one model estimates the overall extraction level of each series, the other the differences between its metals, and the two models together outperform a single model trained directly on the distribution ratios. Used in a screening of ligands, the model identifies the most effective separations (above a factor of two) with an AUC of 0.81, and extractants ranked in the model’s top quartile fall within the true top quartile 50 % of the time, twice the rate expected by chance. The conformal prediction intervals match their nominal coverage. A graph neural network operating on three-dimensional metal–ligand geometries performs as well as the tabular model on its own and adds a small further gain when combined with it.

1 Introduction

Trivalent lanthanides scarcely differ chemically. Along the series, the ionic radius decreases by about 0.013 Å from one lanthanide to the next [1]. A ligand that separates neodymium from praseodymium has, in effect, little more than that shrinkage to work with. An industrial plant therefore runs its extraction stages in long cascades. New extractants are still found, for the most part, by trial and error [2]. Models of the distribution ratio reach R^2 of 0.85 to 0.96 with rows held out at random [2–4], so that a held-out row shares its ligand series with the training set; validated on a known extractant at new acidities, such a model reproduces the level of the series and misses its shape [4], and the shape is the selectivity.

The SAFE database [5] compiles distribution ratios from the solvent-extraction literature, and many of its ligands have been measured across most of the series under a single set of conditions. The separation factor of a pair of metals is the ratio of their distribution ratios from the same experiment, so errors that are common to both measurements cancel. This paper addresses how much adjacent-lanthanide selectivity can be learned from this record for extractants not seen during training, and what a three-dimensional (3D) structure of the metal–ligand complex adds. The work makes the following four contributions: (i) an evaluation score and a model that both separate extraction strength from selectivity ($+0.27 \rightarrow +0.32$ from the model split with the same learner and features); (ii) a small gain from 3D structures of the complexes; (iii) a benchmark of three tight-binding Hamiltonians on the lanthanide contraction, which measures how much information about the metal a computed geometry contains; and (iv) a learning curve, which levels off past a few dozen training extractants.

2 Data and target

We kept the lanthanide(III) rows of the SAFE database that have a resolvable extractant structure, canonicalised the SMILES, removed duplicates and detection-limit placeholders, and retained the 4,746 measurements of $\log D$ on 162 extractants for which a complex geometry exists. Each row carries the extractant, the metal and 64 condition columns. The 956 complexes were built with Architecor [6] (extractant and nitrate or water to coordination number 8–10) and optimised with GFN2-xTB [7].

A separation factor is only defined between measurements made under the same conditions, so rows are grouped into blocks (one extractant, one bin of each of ten condition descriptors). Within a block, replicate $\log D$ measurements of one metal are averaged into a cell, and each pair of cells whose lanthanides lie adjacent in the series contributes one target value,

$$\log \text{SF}_{i,i+1} = \log D_{\text{Ln}_i} - \log D_{\text{Ln}_{i+1}}. \quad (1)$$

Models are trained on $\log D$ row by row and evaluated after this construction is applied to their out-of-fold predictions; the score R_{pair}^2 is the coefficient of determination of predicted against measured $\log \text{SF}$ over all adjacent pairs.

Every number presented below is out-of-fold under five-fold cross-validation grouped by extractant, repeated three times. This yields 905 adjacent pairs on 76 extractants.

3 Model

The tabular representation follows Liu et al. [2]: RDKit descriptors and 2048-bit Morgan fingerprints of the extractant [8], the metal’s atomic number, position in the series and Shannon radius, and the 64 condition columns, 746 in total. The learner is CatBoost [9] with a quantile loss ($\alpha = 0.6$), depth 9 and feature subsampling 0.3, chosen on the adjacent-pair score. Trained on $\log D$ directly, it reaches $R_{\text{pair}}^2 = +0.268$.

The raw target combines the overall extraction strength of a ligand with its variation across metals. Only the variation is scored, so each part receives its own model. The level model f_L is trained on $\log D$, and only its test-block mean, the anchor a_b , is used. The shape model f_S is trained on the block-centred target $\log D_i - \bar{y}_{b(i)}$, where $\bar{y}_{b(i)}$ is the mean measured $\log D$ of the block $b(i)$ that contains row i , and its predictions are centred within the test block in the same way,

$$\hat{y}_i = a_{b(i)} + \left(f_S(x_i) - \text{mean}_{j \in b(i)} f_S(x_j) \right). \quad (2)$$

Because $a_{b(i)}$ is the same for every row of a block, it cancels in every adjacent pair, and R_{pair}^2 depends on the shape model alone. Training that model on the centred target instead of on $\log D$ raises the score from +0.268 to +0.318.

For the 3D channel, heavy atoms of the complex are nodes, atoms within 4 Å are connected, and three continuous-filter layers of width 96 [10] feed a readout with a metal-centred term and a radial shell histogram, since plain mean pooling averages away shifts of a few hundredths of an ångström; the embedding is concatenated with the tabular features, and training uses a Huber loss on $\log D$ plus a contrast term on the within-block pairs of each batch (Appendix A). On its own the network scores +0.266, no better than the boosting baseline. Its block-centred prediction s_{3D} enters Eq. (2) as a second shape source, $\hat{y} = a + (1 - w) s_{\text{tab}} + w s_{3D}$, with $w = 0.35$ chosen on the training folds only; the score becomes +0.326.

4 Results

Why the score is low. Table 1 places R_{pair}^2 on a ladder. The same rows, features and learner give $R^2 = 0.85$ for $\log D$ with rows held out at random, the protocol behind the 0.96 of Liu et al. [4]: the held-out row’s ligand, and usually its neighbours in the series, are in the training set, and the model interpolates. Holding out whole extractants halves this to 0.49. The adjacent-pair target is harder for two further reasons: block means carry 87.5 % of the variance of the cell means and cancel in every pair, so the score is computed on the remaining 12.5 %, and within that part separations measured under independent condition sets agree only to about 0.16 log units against a spread of 0.22. The gains below, from +0.268 to +0.326, are gains on this residual quantity.

The correlation is robust and ligand-specific. On held-out extractants, predicted and measured separation factors correlate with $r = 0.59$ (Figure 1a). A cluster bootstrap over extractants gives an interval of +0.21 to +0.38 for R_{pair}^2 . As a null test, the predictions were shuffled among extractants within each pair position. The shuffled predictions score

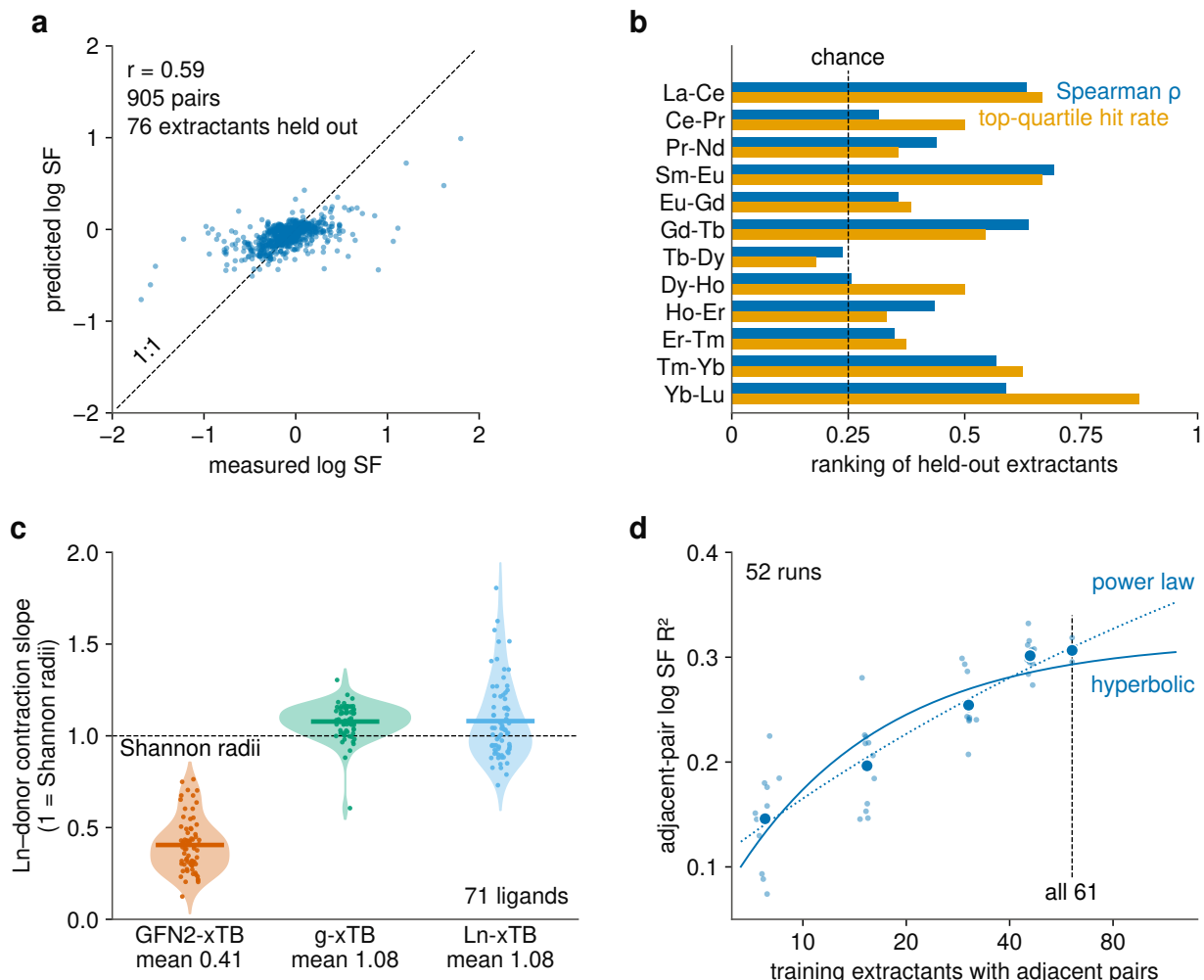


Figure 1: (a) Predicted against measured adjacent-pair log SF for the 905 pairs, extractants held out; the dashed line is the identity. (b) Per adjacent pair: Spearman ρ between predicted and measured extractant rankings, and the top-quartile hit rate against chance. (c) Slope of the Ln-donor distance against the Shannon radius (the compliance c_L) for the 71 ligands under three Hamiltonians: points are ligands, shaded shapes their kernel density, bars the mean; 1 is exact agreement with the radii. (d) Learning curve: small points are the 52 individual runs (four CV seeds, three random draws of the training extractants) at their own training size, large markers the mean per fraction; solid and dotted lines the hyperbolic and power-law extrapolations; dashed line the full training set.

$R^2_{\text{pair}} = -0.05$. Removing both the position means and the extractant means (10 and 13 % of the variance) leaves the ligand-by-position interaction, which is the selectivity itself. On this term the model still scores $+0.23$.

Screening performance. Ten per cent of the pairs account for 71 % of the variance, so R^2_{pair} mostly reflects the large separations. The questions that matter in screening are therefore evaluated directly (Figure 1b). For a given adjacent pair, ranking the held-out extractants gives a mean Spearman correlation of 0.46 over the twelve positions, and the predicted top quartile contains a measured top-quartile ligand 50 % of the time, against 25 % by chance. Used to detect separations with $\text{SF} \geq 2$ (base rate 0.20), the same predictions give an AUC of 0.81. Split-conformal intervals [11], calibrated leave-one-extractant-out, cover 79 % and 90 % of the measured values at nominal levels of 80 and 90 %.

Data requirements. Only seventy-six extractants contribute scored pairs, so a natural concern is that the model is limited by the amount of data. To test this, we trained on a random fraction of the extractants in the training folds

Table 1: One dataset, two splits, two targets. Upper part: the CatBoost model on $\log D$ (Section 3), mean of three single-seed runs. Lower part: R^2_{pair} on the 905 adjacent pairs with extractants held out, seed ensembles. Pair identity alone scores +0.066.

Held out	Model and target	Score
random rows	CatBoost on $\log D$, R^2 of $\log D$	0.85
random rows	same predictions, R^2_{pair}	+0.24
extractants	CatBoost on $\log D$, R^2 of $\log D$	0.49
extractants	same predictions, R^2_{pair}	+0.21
extractants	CatBoost on $\log D$, R^2_{pair}	+0.268
extractants	3D encoder, standalone	+0.266
extractants	Level + shape, Eq. (2)	+0.318
extractants	Level + shape + 3D shape, $w = 0.35$	+0.326

and left the test folds unchanged. R^2_{pair} rises from +0.15 with eight extractants to +0.30 with forty-six, and reaches +0.31 with all sixty-one that a training fold holds, four fifths of the seventy-six (Figure 1d). The last third of the data adds only +0.005, which is within the variation between random seeds, and the two fitted extrapolations place the gain from doubling the number of extractants between +0.01 and +0.04. A few dozen well-measured ligands are therefore sufficient. What is lacking is information that separates ligands within a class (Appendix D).

5 Contribution of three-dimensional structure

The optimised complexes enter the model through the shape term only. Adding the block-centred prediction of the 3D encoder to the tabular shape with weight $w = 0.35$ raises the score from +0.318 to +0.326 (eight CatBoost seeds; 32-seed encoder ensemble), whereas the same geometries converted to 84 descriptor columns and given to a model that predicts $\log D$ directly lower it from +0.278 to -0.022. The gain lies in the light half of the series: on the 482 La–Gd pairs the score rises from +0.356 to +0.369, while on the 347 Tb–Lu pairs it remains at +0.028; the 76 Gd–Tb pairs belong to neither half. The heavy half is where every model has little skill (Spearman 0.30 against 0.58) and where the separations are smaller (SD 0.17 against 0.32).

The gain is limited by the geometries, which vary with the metal through a single number. Eight encoders with different architectures, cutoffs and losses give nearly identical predictions: the eight prediction vectors have an effective rank of 1.05, and predictions from two disjoint eight-seed halves correlate at $r = 0.990$. In GFN2-xTB only Ce and Lu are fitted, and the parameters of the other lanthanides are linear interpolations in Z between them [7], so the optimised metal–donor distances vary smoothly with the position in the series and carry no information about the metal beyond it, which the tabular features already contain. The contraction benchmark (Figure 1c; 71 ligands, fifteen lanthanides substituted into one relaxed complex per ligand and re-optimised with each Hamiltonian) gives a slope of the mean metal–donor distance against the Shannon radius, the compliance c_L (1 is exact agreement with the radii), of 0.405 ± 0.145 for GFN2-xTB, 1.078 ± 0.094 for g-xTB [12], and 1.081 ± 0.214 for Ln-xTB [13] (Appendix C), whose series is irregular (linear-fit R^2 0.43 against 0.82 for g-xTB). Replacing the GFN2-xTB geometries with g-xTB geometries changes the score by +0.005 ($p = 0.68$, eight paired seeds; Appendix B), and the ligand-specific part of the slope is uncorrelated with measured selectivity under all three Hamiltonians ($r = 0.11, -0.02, 0.17$; 44 ligands).

6 Limitations and conclusion

The contribution of the 3D term is smaller than the variation between seeds of a single encoder run. No upper bound on the attainable score can be given: the replicate agreement quoted in Section 4 rests on the 30 % of scored pairs that were measured more than once. Within a single extractant the predicted shape of the series is weak (median $R^2 = 0.09$); the model is useful for deciding which ligands are selective and where their largest separations occur. Within these limits, the published extraction record suffices to predict adjacent-lanthanide selectivity for unseen ligands, and the decisive choice was to predict the shape of a series separately from its level.

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A Protocol

Code, out-of-fold predictions and all results will be released upon publication.

Blocks. The ten binned condition descriptors that define a block are acid class, acid strength, nitrate activity, pH, diluent family, modifier class, temperature, contact time, phase ratio and metal concentration. Pairs with a measured log SF of exactly zero, which arise when both metals sit at a detection limit, are excluded. Promethium is absent, so Nd–Sm is not an adjacent pair.

Hyperparameters. CatBoost: 1,500 iterations, learning rate 0.04, depth 9, ℓ_2 regularisation 3, feature subsampling 0.3, quantile loss $\alpha = 0.6$, identical for the level and shape models. 3D encoder: width 96, three continuous-filter layers, 64 radial basis functions to 5 Å, cutoff 4 Å, heavy atoms only, dropout 0.15, head width 256, Adam at 2×10^{-3} with weight decay 10^{-4} , 60 epochs with early stopping on a validation fold, batches of 64 rows grouped by block so the contrast term always has pairs, pair-contrast weight 4 with adjacent pairs weighted $3\times$. Blend weight: for each fold, selected over $\{0, 0.05, \dots, 1\}$ on the other folds’ out-of-fold pairs, with equal weight per extractant.

Learning curve. Fractions 0.125, 0.25, 0.5, 0.75 and 1 of each fold’s training extractants; within a draw, the subset for a smaller fraction is contained in the subset for the larger one. The full-data run reproduces the stored champion exactly. Because the anchor is one number per block it cancels in every adjacent pair, so the level model does not affect the score and only the shape model’s training set matters.

B Controls

Interventions under the same protocol on the 905 pairs; ΔR^2_{pair} relative to the best comparable system.

Intervention	ΔR^2_{pair}
Simplicial encoder instead of the distance graph	0.00 ($r = 0.96$, blend weight 0.01)
Persistence descriptors as extra shape features	−0.03 to −0.09
Metal-feature weight $\times 2 / \times 5 / \times 10$	+0.00 / −0.05 / −0.08
All-pairs difference model added to the blend	0.00
Condition-transfer features	+0.004
g-xTB geometries instead of GFN2-xTB (802 complexes)	+0.005 ($p = 0.68$)
Complexes rebuilt in correspondence	−0.013
Solvent-relaxed geometries, water / octanol	−0.008 / −0.081
GFN2 / g-xTB interaction-energy features	0.00

C Ln-xTB parameters

The Ln-xTB parameters are given in the supporting information of Zhang [13] as PDF tables only. We parsed both copies (Tables S60–S61 and S62; no disagreement), replaced the $Z = 57\text{--}71$ blocks of the stock GFN2-xTB parameter file, and ran unmodified xtb 6.7.1. Single points on the 35 optimised structures shipped with the paper reproduce the published energies to better than $2 \times 10^{-8} E_h$ once the paper’s spin convention is used (the high-spin count of unpaired f electrons, reduced by one where its parity is inconsistent with the electron count). All 2,130 optimisations of the benchmark converged. A closed-shell control gives $c_L = 0.981 \pm 0.140$ with the same jagged series.

D Americium

SAFE holds 1,794 Am(III) rows on 110 ligands, 95 % of them ligands with lanthanide data, giving 312 Am/Eu separation factors on 82 extractants under identical conditions. Zero-shot, with Am treated as a lanthanide at its radius, the Am/Eu separation is not predicted ($R^2 = -0.14$). Joint training with the Am rows reaches +0.47 on held-out extractants, but a control that assigns each held-out extractant the mean of the other extractants in its ligand family scores +0.47 as well, and within a family the model has no skill (+0.02); the lanthanide score is unchanged ($+0.002 \pm 0.020$ over paired seeds). The model learns which ligand classes discriminate Am from Eu, nothing finer.